NP^3 BLOB LABEL

Usage Notes

https://github.com/danielatrivella/np3_ligand

Cristina Freitas Bazzano

Executing the application:

NP^3 BLOB LABEL

Overview:

https://github.com/danielatrivella/np3_ligand

Files organization

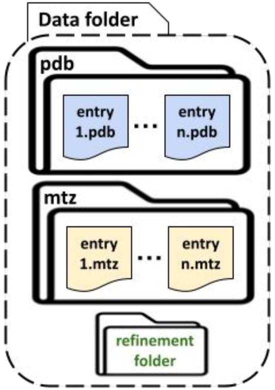
Two cases depending on the entries refinement status

NP³ BLOB LABEL

a) Refine at least one of the entries

Inputs:

1) **Data folder path**
    - **mtz** and **pdb** subfolders
        - One subfolder by entry
    - ***Refinement folder*** *will be created here*
2) **Entries list table path**
    - **Define the entries to refine, otherwise only copy their input to the refinement folder**
3) **Model checkpoint path**
4) **Output path**
5) Blob search parms

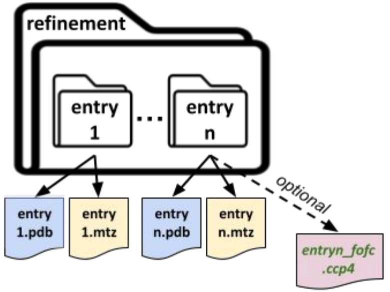


b) Use previous refinement result

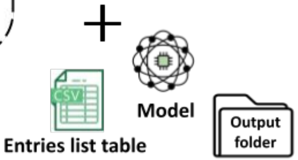
Do not refine the entries (skip step 1)

Inputs:

1) **Entries list table path**
2) **Model checkpoint path**
3) **Output path**
4) **Refinement folder path**
    - One subfolder by entry
    - *Their Fo-Fc map will be exported to a* ***ccp4 map*** *here, if not present yet**
5) Blob search parms



**auto created if not present - useful to use a specific Fo-Fc map, or, e.g., a PanDDA event map*



Input Parameters

Mandatory parameters *in bold*

- ***entries_list_path***: the path to the CSV metadata table defining the entries to be processed (it must be comma separated) - its format is defined in the next slide and depends on the *search_blobs* mode
- ***model_ckpt_path***: the path to a model checkpoint file (provided by the application) - to segment the blob 3D point cloud
    - Model AtomC347CA56
    - Model AtomSymbolGroups
- *search_blobs*: 'all' search for all blobs within the given parameters or 'list' to search only for the blobs listed in specific position in the *entries_list_path* table (default: all)
- *data_folder*: used if *refinement_path* is not informed
    - It must contain the subfolders *pdb* and *mtz* with the entries data; a folder named *refinement_<DATE>* will be created here
- *refinement_path*: path to a previous refinement result, it must contain one subfolder by entry
- *output_name:* Used to name the final result folder in the following format: "*np3_blob_label_<output_name>_<DATE>*".
- *output_path*: Path to the directory where the results will be stored, a subfolder named "*np3_blob_label_<output_name>_<DATE>*" will be created here

- *Additional Find Blobs parameters - defined in the next slide*

Entries List Table - Metadata Format

- The entries list table depends on the Find Blobs (Step 2) mode defined with the parameter *search_blobs*:
    - 'all' -> find all blobs that fulfill the search criteria.
    - 'list' -> Search for blobs in specific positions that fulfill the search criteria.

Find all blobs

entryID	refinement	noHetatm
entry**1**	*1 or 0*	*1 or 0*
…	0 *(disabled)*	0
entry**n**	1 *(enabled)*	1

Search for blobs in specific positions

entryID	refinement	noHetatm	blobID	x	y	z
entry**1**	*1 or 0*	*1 or 0*	blob_1_0	1.1	2.2	3.3
…	0 *(disabled)*	0	…	…	…	…
entry**n**	1 *(enabled)*	1	blob_n_0	0	1	3

- *entryID:* the name of the PDB entries to be processed. It must start with a letter, have unique values and no space or special characters.
    - These values must match the names of the entries input .pdb and .mtz files or refinement subfolders.
- *refinement*: 1 or 0 values defining if the refinement should be executed or not for each entry. If not, the input .pdb and .mtz are copied to the refinement folder (entry already refined). If the *refinement_path* is informed, this columns is not used.
- *noHetatom*: if executing the refinement, this value defines if the heteroatoms should be removed (1) or not (0) before refining the entries.

- *blobID:* name of the blob present in the given position of the given entry. It must start with a letter and have no space or special characters.
    - *entryID* and *blobID* should have unique values when combined.
- x, y and z with the 3D position in Å to search for the blobs in the respective entry.

Additional Find Blobs Parameters

- *sigma_cutoff*: A numeric defining the sigma cutoff to be used to search for blobs in the difference electron density map. Values closer to 3σ are **recommended** (default: 3.0).
    - This parameters is very **memory dependent**, should be used careful depending on your hardware capabilities.
    - Values closer to 2σ may retrieve low quality blobs (which could have a fragmented density) and produce very big point clouds (more than 64Gb of memory RAM is recommended).
    - Values around 2.5σ may retrieve intermediary quality blob, still producing big point clouds (more than 32Gb of RAM is recommended).
    - Values closer to 3σ may retrieve only high quality blobs and produce smaller point clouds (around 12Gb of RAM is enough).
    - Values smaller than 1.5σ are **not recommended**, because they could create a representation of the blob with too much noise, very big and slow to process - high memory footprint (>128Gb of memory RAM may be necessary).
        - This value will directly affect the blobs 3D point cloud creation, and thus, the prediction result.

- *blob_min_volume*: A numeric defining the minimum volume that a blob must have to be considered. Only used when *search_blobs* is 'all'. Default to 24 A^3, what is equivalent to the volume of a water. Smaller values will allow retrieving smaller blobs, the opposite is also true. (default: 24.0)

- *blob_min_score*: A numeric defining the minimum score that a blob must have to be considered. Only used when *search_blobs* is 'all'. The score of a blob is equal to the sum of the difference electron density values of its points. (default: 0)

- *blob_min_peak*: A numeric defining the minimum intensity that the peak (most intense point) of a blob must have for the blob to be considered. Only used when *search_blobs* is 'all'. (default: 0)

Examples from PDB

exemplos_top_down/entries_list_top_down_all.csv

entryID	refinement	noHetatm
4bam	1	1
4xhe	1	1
5f07	1	1
5i8b	1	1
5jtt	1	1
5m8g	1	1
5ybo	1	1
6dir	1	1
4rvn	1	1

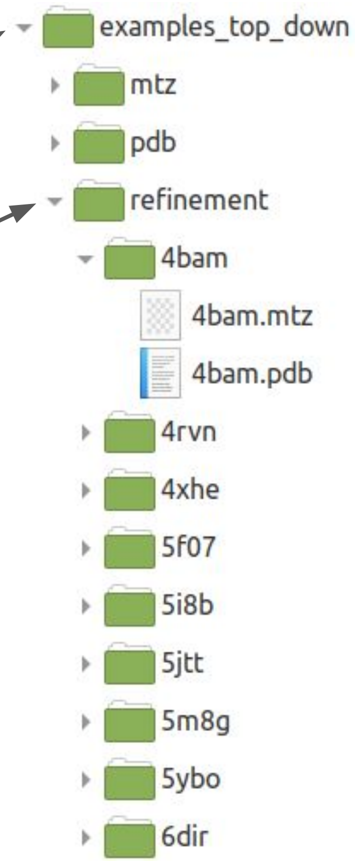
Inputs:

data_folder

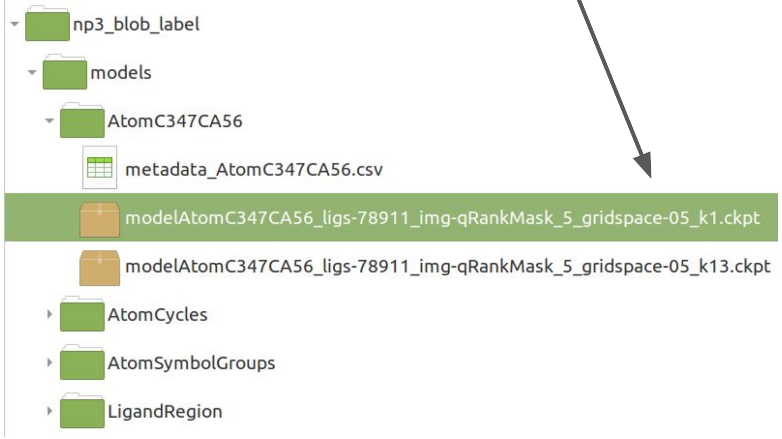
entries_list_path

refinement_path

model_ckpt_path



NP³ Blob Label repository and example:

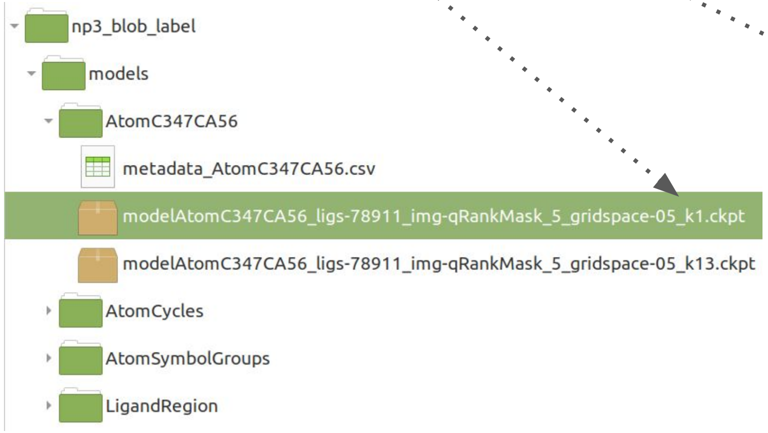


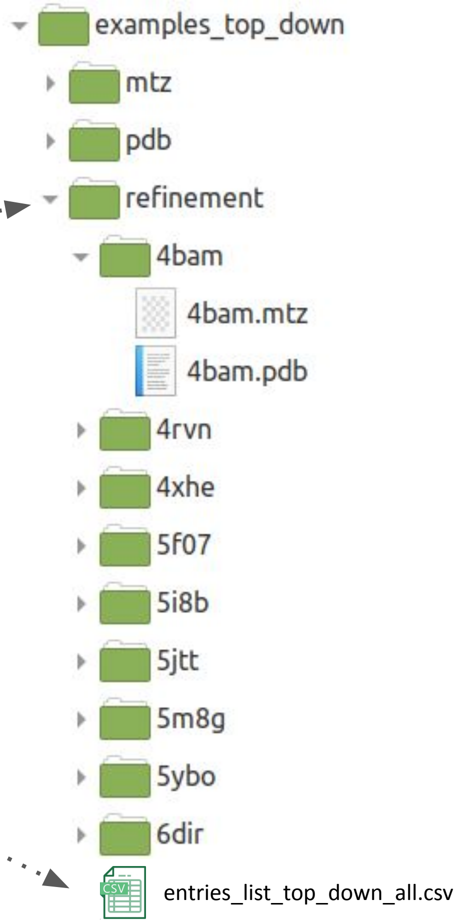
Run - Examples from PDB

- With the requirements installed, open a terminal window inside the NP³ Blob Label repository and run:

```
(np3_blob_label) $ python np3_blob_label.py --data_folder
examples_top_down/ --refinement_path examples_top_down/refinement/
--entries_list_path examples_top_down/entries_list_top_down_all.csv
--model_ckpt_path
models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridsp
ace-05_k1.ckpt --output_name modelAtomC347CA56 --output_path
examples_top_down/
```

NP³ Blob Label repository:

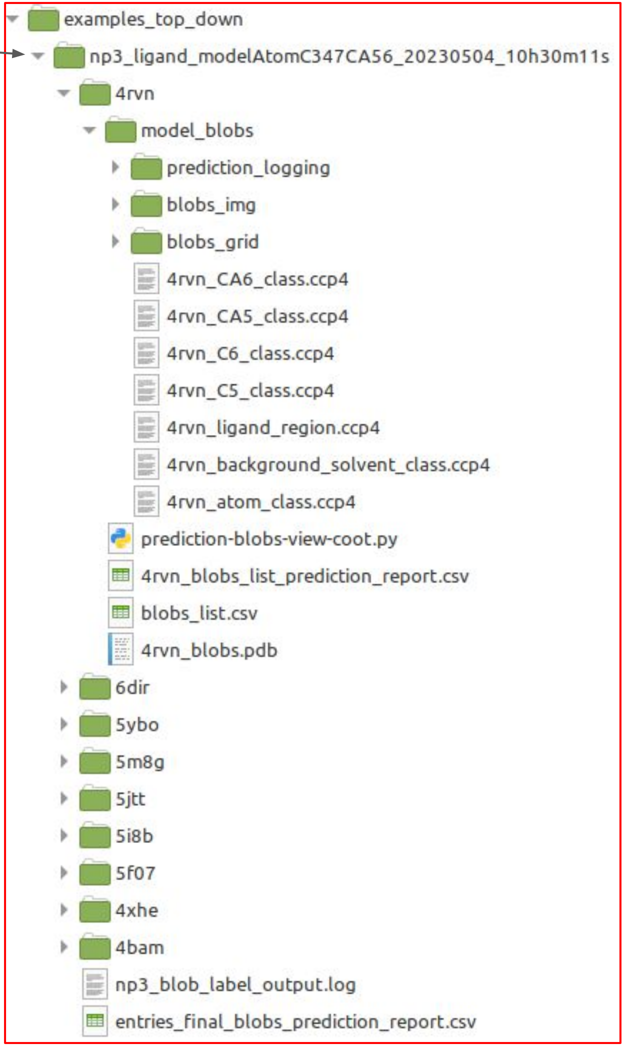




Results - Examples from PDB

The **output** folder is named **"np3_blob_label_<*output_name*>_<Date>"** and is stored inside the *data_folder* path. It contains:

- One subfolder by entry, named with the "entryID", containing:
  - The blobs' modeling result in the "model_blobs" folder, containing:
    - The blobs' grid ("blobs_grid" folder) and final point clouds ("blobs_img" folder)
    - The prediction logging file ("prediciton_logging" folder)
    - The predicted classes in CCP4 maps:
      - One map by predicted class named: "*<entryID>_<predicted_class>_class.ccp4*"
      - One map with all predicted points except background (ligand region): "*<entryID>_ligand_region.ccp4*”
  - A Coot script to automatically visualize the results: "prediction-blobs-view-coot.py"
  - A table listing the blobs that were found named "blobs_list.csv" and another table with the respective entry report, listing the blobs and their predictions result, named "<entryID>_blobs_list_prediction_report.csv"
  - A "<entryID>_blobs.pdb" file with fake atoms placed in the blobs center position
- One Logging file named "np3_blob_label_output.log"
- One global report file named "entries_final_blobs_prediction_report.csv"



Output Report Table - Examples from PDB

At the end of the process, the user may inspect all the found blobs of the given entries in the final report table:

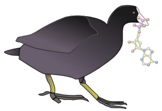
- File np3_blob_label_<output_name>_<Date>/entries_final_blobs_prediction_report.csv

The final report table contains one found blob by row with their information by column, such as the blobs' intensity, volume, score and position, and their predicted classes by size (number of labeled points in each class). This table may help the user summarize the findings and prioritize further analysis.

The description of the columns of this table are presented in the file 'np3_blob_label/docs/report_table_columns_description.csv'

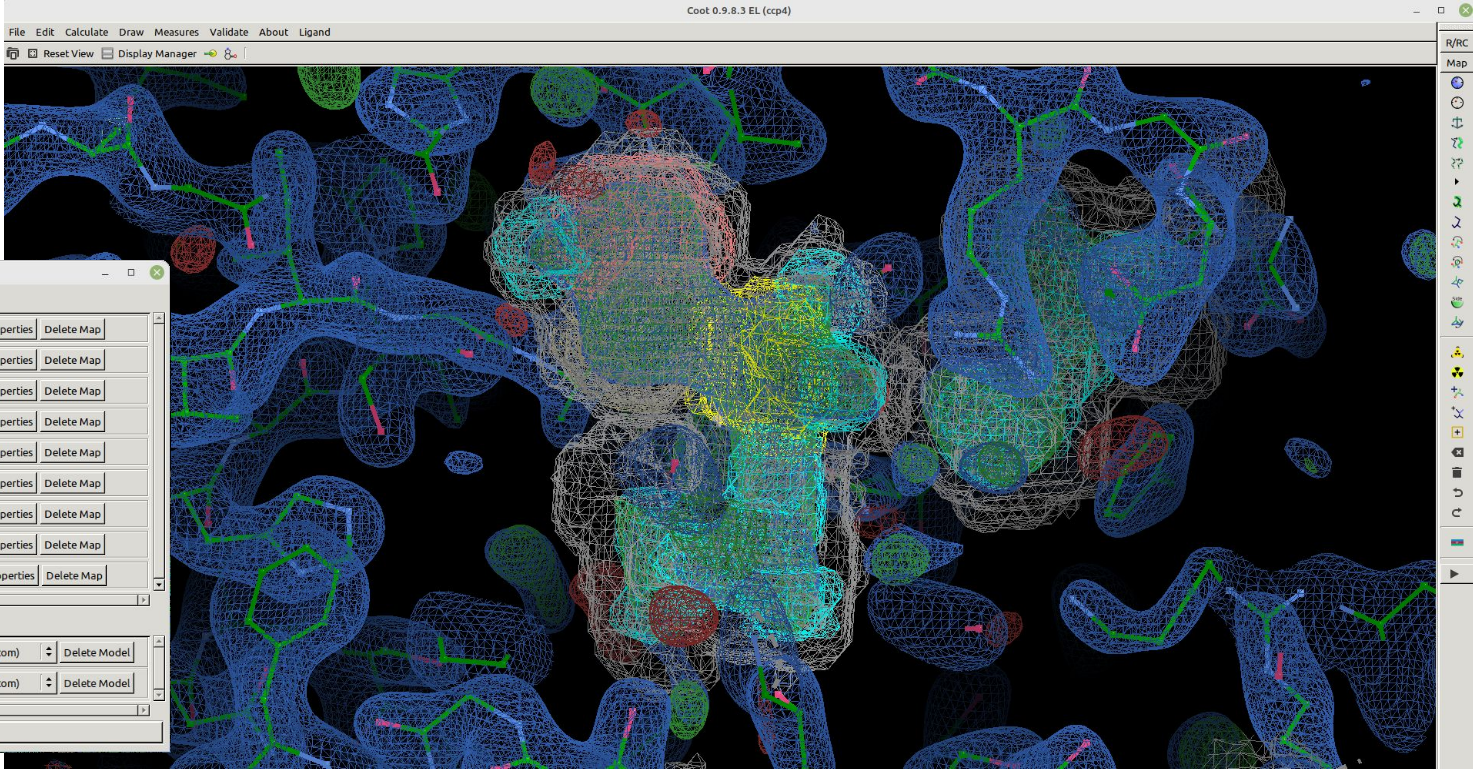
Output Visualization in Coot - Examples from PDB

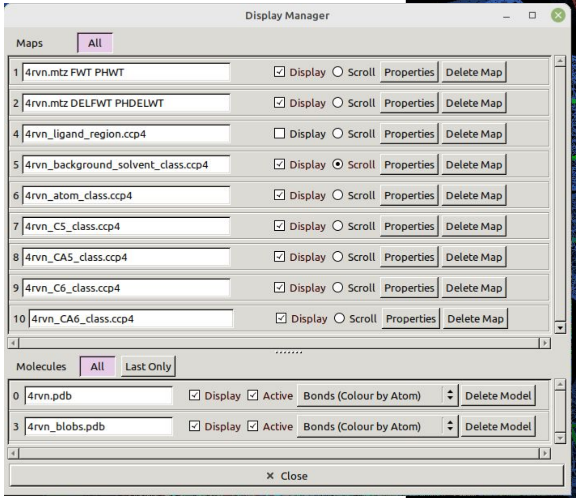
- To execute the NP³ Blob Label Coot script for entry **4rvn**, open a terminal window and run:



```
$ coot --script examples_top_down/np3_blob_label_modelAtomC347CA56_<Date>/4rvn/prediction-blobs-view-coot.py
```

Only the map with the ligand region is not displayed in the beginning



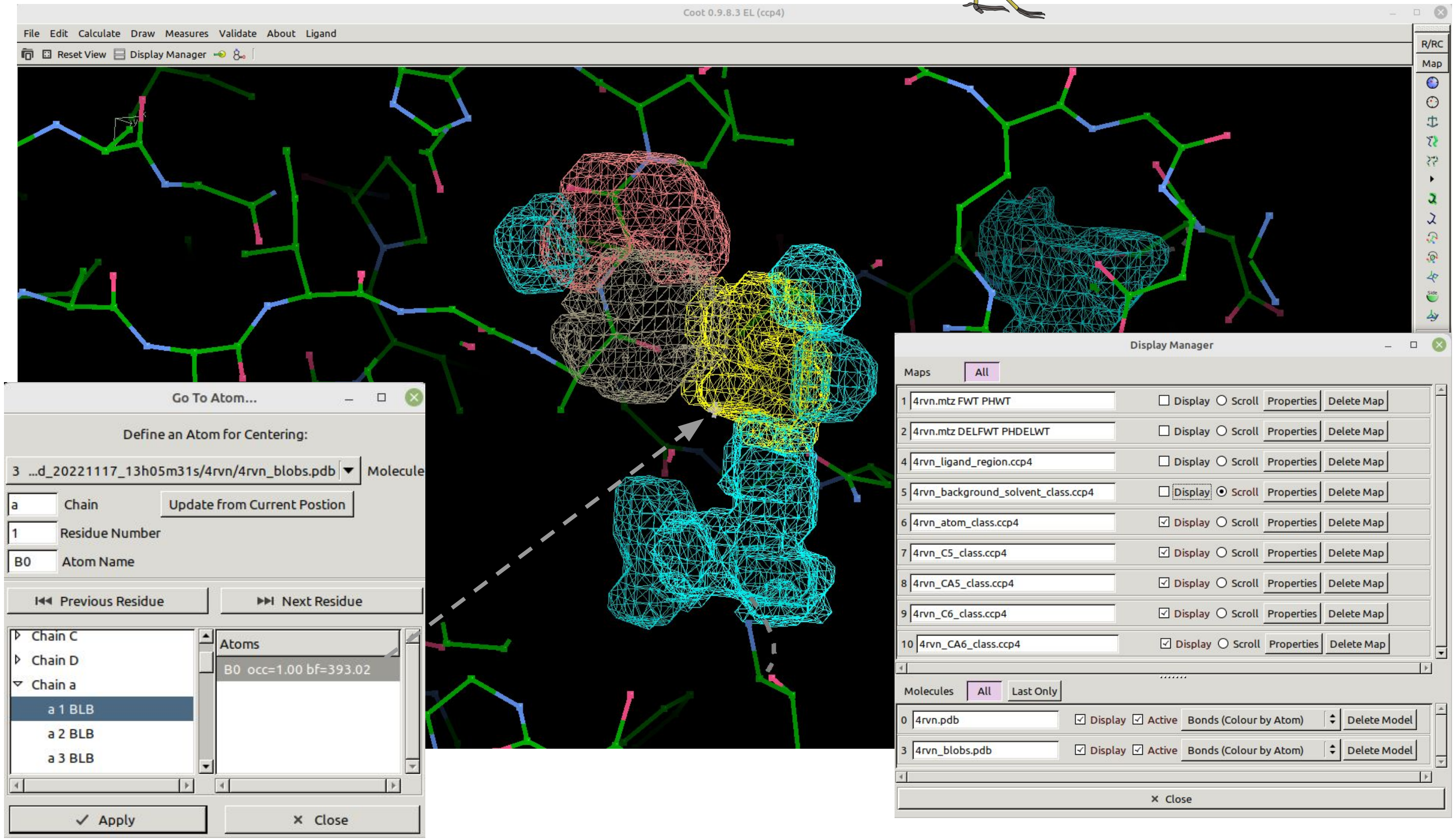


Output Visualization in Coot - Examples from PDB

For a better visualization of the result the user may set off the display of the experimental electron density maps and the background class map

The background class may help visualize the format of the 3D point cloud of the blobs

The Coot "Go To Atom…" tool may help navigate through the result, by centering in each found blob present in the last chain



NP³ Blob Label Commands - Examples

- Run NP³ Blob Label using a previous refinement result to search for all blobs in the given entries:

```
$ python np3_blob_label.py  --entries_list_path examples_top_down/entries_list_top_down_all.csv   --refinement_path examples_top_down/refinement/  --model_ckpt_path models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt --output_path examples_top_down/  --output_name all_blobs_modelAtomC347CA56
```

- Run NP³ Blob Label to refine the entries specified in the entries list table and to search for blobs in specific positions:

```
$ python np3_blob_label.py  --data_folder examples_top_down/  --entries_list_path examples_top_down/entries_list_top_down_list.csv --model_ckpt_path models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt    --search_blobs list --output_path examples_top_down/  --output_name list_blobs_modelAtomC347CA56
```

- Run NP³ Blob Label using a previous refinement result to search for all blobs in the given entries and to use the model AtomSymbolGroups to segment the blobs:

```
$ python np3_blob_label.py  --entries_list_path examples_top_down/entries_list_top_down_all.csv   --refinement_path examples_top_down/refinement/  --model_ckpt_path models/AtomSymbolGroups/modelAtomSymbolGroups_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt    --output_path examples_top_down/ --output_name all_blobs_modelAtomSymbolGroups
```

- Check the list of parameters with their full description, default values and expected values for NP³ Blob Label. Mandatory parameters have a default value equal to None:

```
$ python np3_blob_label.py  --help
```

NP³ Blob Label Commands - Examples

- Run NP³ Blob Label using a previous refinement result to search for all blobs in the given entries with sigma cutoff equal to 2 (high memory required):

```
$ python np3_blob_label.py  --entries_list_path examples_top_down/entries_list_top_down_all.csv   --refinement_path
examples_top_down/refinement/  --model_ckpt_path models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt
--output_path examples_top_down/  --output_name all_blobs_modelAtomC347CA56_2sigma  --sigma_cutoff 2
```

- Run NP³ Blob Label using a previous refinement result to search for all blobs in the given entries with sigma cutoff equal to 2 σ and minimum blob volume equal to 50 Å³:

```
$ python np3_blob_label.py  --entries_list_path examples_top_down/entries_list_top_down_all.csv   --refinement_path
examples_top_down/refinement/  --model_ckpt_path models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt
--output_path examples_top_down/  --output_name all_blobs_modelAtomC347CA56_2sigma_minV_50  --sigma_cutoff 2 --blob_min_volume 50
```

- Run NP³ Blob Label using a previous refinement result to search for all blobs in the given entries with sigma cutoff equal to 2.5 σ and number of CPU cores for parallelization equal to 4 (very high memory required):

```
$ python np3_blob_label.py  --entries_list_path examples_top_down/entries_list_top_down_all.csv   --refinement_path
examples_top_down/refinement/  --model_ckpt_path models/AtomC347CA56/modelAtomC347CA56_ligs-78911_img-qRankMask_5_gridspace-05_k1.ckpt
--output_path examples_top_down/  --output_name all_blobs_modelAtomC347CA56_2.5sigma  --sigma_cutoff 2.5 --parallel_cores 4
```

NP³ BLOB LABEL

Get in contact with the NP³ team
for more information about the NP³ Blob Label application

- Open an issue/discussion in the github repository:
  - https://github.com/danielatrivella/np3_ligand/issues
- Send an email with your question or comment:
  - *cristina.freitas@lnbio.cnpem.br*
  - *daniela.trivella@lnbio.cnpem.br*

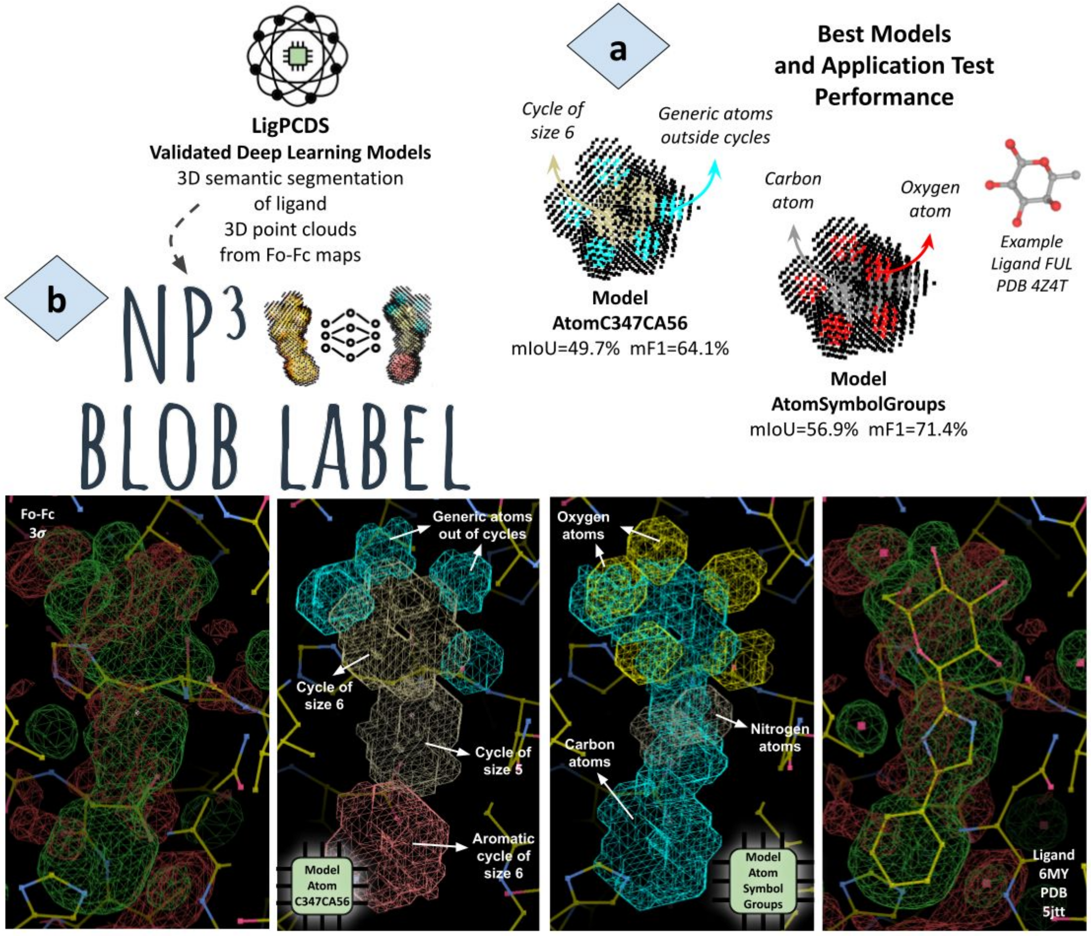
Thanks!

Summary

The NP³ Blob Label is a deep learning semi-automatic solution to auxiliate in ligand building tasks.

It is capable of searching for new ligands sites in difference electron density maps, called blobs, and for each found blob the application will use the validated DL models from LigPCDS (a) to predict chemical substructures that fill and explain each part of the blob.

The predictions are converted to CCP4 maps for the visualization of the results and serve as an initial proposal to help in the complete manual reconstructions of the ligand chemical structure, as illustrated in (b).



3D Image of the difference electron density of a ligand *Blob*

Blob labeled with generic atoms and cyclic substructures

Blob labeled with atoms symbols

Ligand chemical structure manually built

How

The deep learning models were trained in semantic segmentation tasks and validated with a dataset of **78902** ligands 3D point clouds. These representations are from the difference electron density (Fo-Fc) map of **26976** PDB entries and **11925** unique ligands in a resolution range between **1.5 Å** and **2.2 Å**. Data outside this resolution range may still be used in the NP³ Blob Label application, but the models accuracy (presented next) may not be reliable.

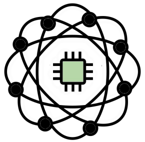
The application will create a 3D point cloud for each found blobs from its Fo-Fc map and label this representation using the models prediction result. The blob point cloud will be slight affected by the σ contour level used in the search and the labels depends on the model used for the predictions. A schema of the application pipeline is illustrated next.

The following models were validated by LigPCDS and gave the best result. They are available in the NP³ Blob Label repository (*models* folder):

Vocabulary
Background
Atom
C5
CA5
C6
CA6
C3
C4
C7

Model AtomC347CA56

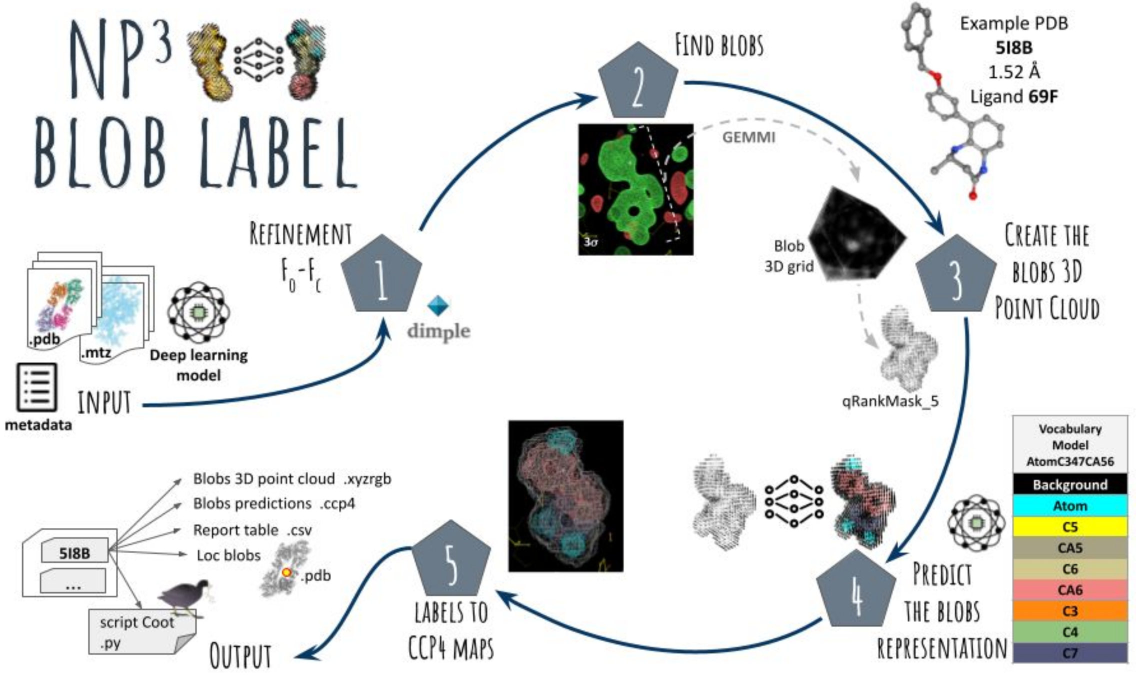
Predicts generic atoms out of cycles, cycles with sizes from 3 to 7 and aromatic cycles with sizes 5 and 6



Model AtomSymbolGroups

Predicts the atoms symbols with groupings. The halo group contains the halogen atoms (Cl,Br,I,F) and the PSe group contain the minority atoms (P,S,Se)

Vocabulary
Background
C
O
N
PSe
Halo



The **NP³ Blob Label** is fully automated in python scripts and its functionality was wrapped up in a command line interface with few inputs. The code is freely available in the github repository: *https://github.com/danielatrivella/np3_ligand*

The NP³ Blob Label **pipeline workflow:**

(1) Refine the entries with Dimple;

(2) Search for blobs in the entire calculated Fo-Fc map or in given specific positions

- Uses a σ contour level for the search and other parameters;
- Limit the blob grid bounds to 50Å

(3) Create the final representation of the blobs in 3D point cloud (qRankMask_5);

- Same methodology used by LigPCDS

(4) Label these representations using a validated DL model prediction; and

(5) Convert the prediction result to synthetic electron density maps in CCP4 format, with each segmented class in a different map.

At the end of the workflow the user may easily visualize the result of each entry with the Python script created for Coot, which automatically opens the inputs (.mtz and .pdb) of an entry along with the synthetic electron density maps of each segmented class of the found blobs. The user may **browse the found blobs** and visualize their predictions using Coot's atom navigation tool (using the .pdb file of the protein entry with dummy atoms centered at the position of each found blob). The application result also contains a **report table** with all found blobs, their information (intensity, volume, score and position) and their predicted classes by size (number of labeled points in each class), which may help the user summarize the findings and prioritize further analysis.

Best Models Accuracy

Three different values for the σ contour level parameter were used to evaluate the impact of this parameter on the accuracy of the application. Contours equal to 2, 2.5 and 3σ were used in the blobs search. For the other parameters, their default values were maintained (minimum volume set to 24 $Å^3$ and minimum score and peak intensity set to zero).

The results of these tests are presented below with their accuracies in terms of mIoU, mF1, Precision and Recall and the percentage of entries that were correctly imaged (3D point cloud created).

Results from the hold-out CV against 3036 ligands from the k=1 test subset of the stratified training dataset from LigPCDS for model AtomSymbolGroups and against 3035 from k=13 test subset for model AtomC347CA56. Best values by model are highlighted.

DL Model	Sigma Contour Level	Number of Blobs Imaged	Test mIoU	F1 score	Precision	Recall
AtomC347CA56	2	**2674 (88%)**	47.4	61.8	61	64.1
AtomC347CA56	2.5	2523 (83%)	48	62.2	60.7	65.2
AtomC347CA56	**3**	2362 (78%)	**49.7**	**64.1**	**61.3**	**69.4**
AtomSymbolGroups	2	**2687 (89%)**	54.4	69.2	67.3	72.9
AtomSymbolGroups	2.5	2540 (84%)	56	70.6	68.2	74.7
AtomSymbolGroups	**3**	2367 (78%)	**56.9**	**71.4**	**68.4**	**76.2**

Best Models
Confusion Matrices

Model AtomC347CA56

3 σ	Background	Atom	C5	CA5	C6	CA6	C3	C4	C7
Background	**86.5**	3.8	0.3	0.1	2.0	0.6	0.0	0.0	0.0
Atom	21.5	**54.3**	0.4	0.0	1.1	0.6	0.0	0.0	0.0
C5	13.7	2.8	**59.9**	1.2	2.0	1.3	0.0	0.0	0.0
CA5	14.0	2.5	2.7	**64.7**	0.8	6.1	0.0	0.0	0.0
C6	13.7	2.8	0.6	0.1	**36.1**	3.7	0.0	0.0	0.0
CA6	15.5	2.5	0.4	0.3	3.2	**60.7**	0.0	0.0	0.0
C3	20.3	22.1	0.0	0.0	1.1	0.3	**22.9**	1.5	0.0
C4	23.3	10.0	13.8	5.4	0.2	1.1	0.0	**30.3**	0.0
C7	19.7	4.3	0.2	1.5	26.7	0.3	0.0	0.0	**31.6**

mIoU = 49.7%

Model AtomSymbolGroups

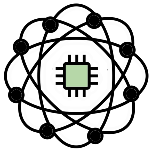
Predicted Class

Expected Class

3 σ	Background	C	O	N	PSe	Halo
Background	**86.7**	4.3	2.0	0.3	0.1	0.0
C	17.3	**54.8**	1.4	0.7	0.1	0.0
O	19.8	6.3	**48.3**	0.8	0.5	0.1
N	16.6	16.4	3.5	**49.4**	0.1	0.0
PSe	11.3	3.7	3.9	0.1	**61.8**	0.2
Halo	23.5	11.5	10.3	0.6	0.6	**40.2**

mIoU = 56.9%

The diagonal of the confusion matrix above shows the accuracy by class in terms of Intersection over Union (IoU) from 0 to 100%.



The mean IoU (mIoU) is the accuracy for the entire model. The best σ contour level was used equal to 3σ.

The confusion matrix have the expected Class in the row and the predicted Class in the column. They were normalized by row.